

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

CRUDE DISTILLATION UNIT (CDU-1) - PERIODIC EQUIPMENT INSPECTION LOG

Equipment ID: Furnace F-101	Inspection Date: 2026-08-24
Service: Crude Charge Pre-Heating	Shift / Lead: Night Shift B / Lead Insp. R. Sharma
Operating Standard: SOP-MRPL-FURNACE-01	Report ID: MRPL-INS-CDU1-2026-0894

1. Radiant Tube Skin Thermocouple Telemetry & Integrity Readings

Sensor Tag	Measurement Location	Recorded Temp	Design Limit	Status / Verdict
TT-101	Radiant Section - Zone A Inlet	582 °C	610 °C	NORMAL (Compliant)
TT-102	Radiant Section - Zone A Midpoint	594 °C	610 °C	NORMAL (Compliant)
TT-103	Radiant Section - Zone B Lower	604 °C	610 °C	ELEVATED (Monitor)
TT-104	Radiant Section - Zone B Upper	620 °C	610 °C	CRITICAL BREACH (>610°C)
TT-105	Convection Bank Inlet	415 °C	450 °C	NORMAL (Compliant)
TT-106	Convection Bank Exit	385 °C	410 °C	NORMAL (Compliant)

2. Metallurgical Wall Thickness & Firing Conditions

Parameter	Measured Value	Standard Threshold	Action Directive
Wall Thinning / Corrosion Rate	0.45 mm/year	Max 0.25 mm/year	Exceeds Corrosion Allowance
Ultrasonic Residual Thickness	4.62 mm	Min 5.00 mm (SOP 4.1.3)	Retire / Sleeve Required
Burner Firing Rate	104% MCR	100% Maximum Continuous	De-rate Burners to 90% MCR
Excess O2 Flue Gas Level	1.8 vol%	2.5 - 3.0 vol%	Adjust Draft Damper

3. Lead Inspector Engineering Findings & Mandatory Action

FINDING: Radiant coil thermocouple TE-104 has consistently logged skin temperatures at **620 °C**, exceeding the design envelope limit of **610 °C** stipulated in SOP-MRPL-FURNACE-01 Clause 4.1.2. Ultrasonic measurement indicates severe localized wall loss (residual 4.62 mm vs 5.0 mm allowable minimum).

DIRECTIVE: In accordance with MRPL Refinery Operating Standard Clause 4.1.2, mandatory load de-rating to 75% capacity is required immediately, followed by emergency furnace turnaround and decoking within **7 days**.

Inspected By: R. Sharma, Lead NDT Inspector (MRPL)	Reviewed By: Chief General Manager (Operations)
Status: ESCALATED TO CRITICAL AUDIT	Air-Gap Cryptographic Hash: SEALED